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Spread-Spectrum Method Using Multiple Sequences for Underwater Acoustic Communications

Fengzhong Qu, *Senior Member, IEEE*, Xiangzhao Qin, Liuqing Yang , *Fellow, IEEE*, and T. C. Yang

Abstract—Direct-sequence spread-spectrum-based approaches have been adopted in underwater acoustic communications. In this paper, multiple sequences utilizing the periodic correlation characteristics of an original spreading waveform are superimposed, modulating multiple distinct symbols simultaneously within each block at the transmitter. Among those superimposed symbols, one is used as the pilot for channel estimation on chip level and others carry data. Similar to existing superposition schemes, the proposed approach also has low receiver complexity, only requiring matched filtering operations. However, the innovation of the proposed approach is perfect interference suppression, cyclic prefix (CP) is inserted to eliminate the interblock interference totally. Combined CP operation leading to an equivalent cyclic channel matrix with circularly shifted versions of spreading sequences, interferences can be solved considerably. The proposed technique is evaluated with both simulation and experimental data analysis. Data processing gives a brief discussion about low probability of detection and low probability of interception.

Index Terms—Direct-sequence spread spectrum (DSSS), interference suppression, low probability of detection (LPD) and low probability of interception (LPI), multiple sequences, periodic correlations.

I. INTRODUCTION

THE need for high reliability underwater wireless communications arises in military, scientific, and civilian applications, involving submarine communications, scientific data collections, and data exchanges via the underwater sensor networks, speech transmissions between divers and offshore oil explorations, etc., [1]. Meanwhile, achieving reliable communications over underwater acoustic (UWA) channels is recognized as a challenge owing to the scarce bandwidth available and the severely double spreading phenomenon both in the time (multipath delay spread) and frequency (Doppler spread) domains [2]. In addition, autonomous underwater vehicles (AUVs) and sensor receivers have limited signal processing capability, only accommodating very few hydrophones due to the size and power limitations. All

the obstacles mentioned above make it difficult to extract the desired signals from the incoming measurements.

Direct-sequence spread-spectrum (DSSS)-based techniques are adopted for their robustness at low signal-to-noise ratio (SNR). Many works have been done extensively from suitable modulation to reception design [3]–[19]. A novel double-differentially coded spread-spectrum system structure for long-range communications is proposed in [3], double-differentially coding makes it possible to recover signal without explicit phase, Doppler tracking and correction in long range UWA communications. He *et al.* [4] gives a variable data rate based DSSS scheme, involving complementary code keying for shallow-water acoustic channels. The major obstacles encountered on UWA channels are extended multipath propagation and rapid time variation, causing severe time spreading, and frequency-selective fading; accurate channel estimation can enhance system performance, as in [5]–[8]. Considering both channel delay and Doppler parameters jointly, Iltis and Fuxjaeger [5] applies a coherent RAKE receiver coupled with an extended Kalman-filter based estimator for the channel parameters. UWA channels are sparse both in delay and Doppler degree, by which motivation, the orthogonal pursuit algorithm sparse channel estimation is performed on chip level in [8], coupled with a two-dimensional grid of path delay and Doppler scaling factor incorporated in the dictionary construction. Many approaches have been presented in the literature on the receiver design for DSSS or CDMA systems. By combining energy on multipath, RAKE receiver is a good choice in DSSS UWA communications to collect multipath diversity [8], [9]. Meanwhile, decision feedback equalizer (DFE) based receivers have been widely adopted in phase-coherent communications to eliminate severe inter-chip and intersymbol interferences. Inspired by this, DFE receivers are applied on symbol or chip level to suppress multipath interferences, with despreading operation collecting spreading gain in [7], [10], [11]. A comprehensive performance comparison of RAKE and hypothesis feedback DSSS techniques is given in [12]. High SNRs are often required to achieve symbol synchronization and channel equalization for the mentioned methods above. Low probability of detection (LPD) is considered in [13]–[15], achieving high processing gain by taking the time-updated channel impulse response (CIR) as matched filter to eliminate multipath spread.

Some proposed methods in [20] and [21] utilize a set of M-ary pseudonoise (PN) sequences with orthogonality superimposed, transmitting binary message in parallel, e.g., binary phase-shift keying (BPSK) and differential binary phase-shift keying (DBPSK); the receiver consists of the same sets of waveforms as the matched filters to demodulate. Since the matched-filter despreading is based on the aperiodic correlation characteristics of PN sequences, interferences cannot be solved perfectly. Different from the schemes mentioned above, in this paper, the superimposed sequences are acquired by shifting one original sequence with good periodic correlations circularly, transmit-

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F. Qu and X. Qin are with the Ocean College, Zhejiang University, Zhoushan 316000, China (e-mail: jimqufz@zju.edu.cn; chrisqinxz@zju.edu.cn).

L. Yang is with the Department of Electrical and Computer Engineering, Colorado State University, Fort Collins, CO 80523 USA (e-mail: lqyang@engr.colostate.edu).

T. C. Yang is with the Department of Electrical and Computer Engineering, Zhejiang University, Hangzhou 310058, China (e-mail: tsihyuang@gmail.com). Digital Object Identifier 10.1109/JOE.2017.2750298

ting multiple M-ary phase-shift keying (PSK) symbols in superimposition. At the receiver, the same circularly shifted versions of the original spreading sequence are taken as the matched filter to despread the received signals individually, realizing perfect interference suppression in theory. Among the despreading streams, one is taken as the pilot for channel estimation, then the estimated channel serves as matched filter again to collect multipath diversity. The contributions of this paper are summarized as follows.

- 1) Different from existing literature on superposition transmission, perfect interference suppression is more emphasized. Combined CP operation leading to an equivalent cyclic channel matrix with circularly shifted versions of an original spreading sequence, interblock interference (IBI) caused by multipath delay and cosymbol interferences from superimposed symbols transmitted simultaneously can be solved perfectly in theory.
- 2) A generalized structure of PN sequences with good periodic autocorrelation properties is utilized, two types of which, m-sequence and binary zero correlation zone (ZCZ) sequence proposed in [22] are analyzed, especially the latter one whose interferences over certain time lags are zero.
- 3) A closed form of signal-to-interference ratio (SIR) is given as a performance analysis in this paper, proving that the proposed approach can achieve interference suppression considerably.
- 4) Experimental results give a brief discussion about LPD and LPI with the proposed scheme.

The proposed scheme is compared with a conversional decision-directed (D-D) channel estimation based DSSS scheme via simulations. Numerical results show that the proposed one outperforms the D-D DSSS scheme superiorly. Sea trial data sets are acquired from the Acoustic Communication Experiment 2015 implemented in the area of Zhoushan, Zhejiang, China. Data processing results show that ZCZ sequence superimposed exhibits better in the sense of suppressing inter- and cosymbol interferences, the average bit error rate (BER) is less than 1% for input-SNR ≥ -10 dB, and all packets are error free when input-SNR ≥ -6 dB.

The paper is structured as follows. Section II elaborates the proposed system model. Simulations and experimental results are then presented in Sections III and IV, respectively. Finally, concluding remarks are given in Section V.

Notation: Upper (lower) boldface letters are used for matrices (column vectors). $(\cdot)^H$, $(\cdot)^T$, and $(\cdot)^*$ represent the Hermitian, transposition and conjugation, respectively. $\mathbf{I}_N$ and $\mathbf{0}_{N \times M}$ denote the $N \times N$ identity matrix and the $N \times M$ all 0 matrix. $\lfloor \cdot \rfloor$ denotes the floor, $\|\cdot\|^2$ is the squared Frobenius norm of a matrix or vector, $\|\cdot\|$ is the norm of a scalar, $\mathbb{E}\{\cdot\}$ represents the expected value, $[\mathbf{A}]_{i,j}$ denotes the element on the i th row, and the j th column of matrix $\mathbf{A}$.

II. SYSTEM MODEL

A. Transmitted Signals

In traditional DSSS systems, a single symbol is modulated per spreading sequence. In a vector form, the i th modulated signal block is

$$\mathbf{x}(i) = s(i)\mathbf{c} \quad (1)$$

where $s(i)$ represents the i th transmitted symbol and $\mathbf{c}$ is the spreading sequence. The spreading code is the maximum length sequence (m-sequence) in [14].

Differing from the method in [14], the proposed approach adopts a set of generalized spreading sequences with periodic autocorrelation characteristics, e.g., m-sequence and ZCZ sequence. Furthermore, instead of transmitting one symbol per block as in (1), the proposed

scheme modulates multiple distinct symbols (PSK) on superimposed shift sequences during each block, increasing the data rate. Although the payload symbol is quadrature phase-shift keying (QPSK) modulation scheme only, derivation process provided in the following sections can be easily extended to a general M-ary PSK case.

Define the circular shift matrix as

$$\mathbf{T} := \begin{bmatrix} \mathbf{0}_{1 \times (M-1)} & 1 \\ \mathbf{I}_{M-1} & \mathbf{0}_{(M-1) \times 1} \end{bmatrix} \quad (2)$$

which introduces a circular shift by 1 upon premultiplying an $(M \times 1)$ vector. Accordingly, vector $\mathbf{T}^j \mathbf{c}$ is the form of an original spreading sequence $\mathbf{c}$ circularly shifted by j chips. Note that an m-sequence and its circularly shifted version have the following auto-correlation property

$$\mathbf{c}^T \mathbf{T}^m \mathbf{c} = \begin{cases} M, & (m \bmod M) = 0 \\ -1, & \text{otherwise.} \end{cases} \quad (3)$$

Hence, distinct signals through UWA channels riding on $\mathbf{c}$ and $\mathbf{T}^m \mathbf{c}$ will induce negligible interferences among themselves, as long as the circular shift $m \geq 1$. An ideal spreading waveform whose periodic correlations over certain lags are zero can efficiently suppress inter- and cosymbol interferences, realizing interference-free transmission; the features of m-sequences as well as ZCZ sequences will be discussed in the sequel. Nevertheless, UWA channels are well known to have extensive multipath. Let $\tau_{\max}$ denote the maximum delay spread and T_c is the chip duration. The multipath essentially spreads over $(L + 1)$ chips, where

$$L := \lfloor \tau_{\max} / T_c \rfloor. \quad (4)$$

To separate the delayed multipath components of superimposed symbols sufficiently, the circular shift span between the sequences conveying superimposed symbols should be at least $(L + 1)$ chips. The transmitted signal block in our proposed scheme is given by

$$\mathbf{x}(i) = \sum_{j=0}^{J-1} s(i; j) \mathbf{T}^{j(L+1)} \mathbf{c} \quad (5)$$

where $s(i; j)$ is the j th transmitted symbol during the i th block, and J is the number of superimposed sequences per block. J is upper bounded by $J_{\max}$

$$J_{\max} = \left\lfloor \frac{M}{L+1} \right\rfloor \quad (6)$$

where M is the periodic correlation length of the spreading sequence, equating sequence length. It is also worth mentioning that $J_{\max}$ is the maximum number of symbols that can be simultaneously transmitted when the distribution and strength of the actual (and possibly sparse) channel taps are not available at the transmitter. If this information is available, then it is possible to increase $J_{\max}$ by smartly scheduling the signals. In addition, in a very low rate system, one can also choose to transmit J ($2 \leq J \leq J_{\max}$) symbols to further reduce the intersymbol interferences. In this design, the only information *a priori* about the channel needed at the transmitter is the multipath delay spread or an upper bound on it.

B. UWA Channel Propagation

The statistical characteristics in UWA channels remain stable from about hundreds of milliseconds to tens of minutes as shown in [23]. For a typical UWA DSSS system with 10 kilochips per second (kc/s), adopting a spreading sequence consisting of 1000 chips, the sequence block duration is 100 ms. Hence, it is reasonable to assume that the

channel remains time-invariant within a sequence block and is allowed to change across blocks in a time-variant channel.

For block-wise transmissions over multipath channels, the i th $(M \times 1)$ received block $\mathbf{r}(i)$ contains not only the signals from the i th transmitted block, but also the IBI from the previous block. The received signal in vector form can be written as

$$\mathbf{r}(i) = \mathbf{H}(i)\mathbf{x}(i) + \mathbf{H}_{\text{IBI}}(i)\mathbf{x}(i-1) + \mathbf{z}(i) \quad (7)$$

where the second term $\mathbf{H}_{\text{IBI}}(i)\mathbf{x}(i-1)$ is the IBI, the $(M \times M)$ channel matrices are given by $[\mathbf{H}(i)]_{m,n} = h(i; m-n)$ and $[\mathbf{H}_{\text{IBI}}(i)]_{m,n} = h(i; M+m-n)$, for $m, n = 1, \dots, M$, and $\mathbf{z}(i)$ represents the additive white Gaussian noise (AWGN) taking account of thermal or hardware related noise, interferences or jamming, as well as the overwhelming ambient noise. Each element of $\mathbf{z}(i)$ is assumed to be a circularly symmetric independent and identically distributed (i.i.d.) complex-valued Gaussian random process, with zero mean and variance σ^2 denoted as $\mathbf{z}(i) \sim \mathcal{CN}(0, \sigma^2 \mathbf{I})$ in [9].

To eliminate the IBI in (7), at least two remedies are available [24]. One inserts a CP with length L to each block at the transmitter, and removes the CP at the receiver. The equivalent channel after CP insertion and removal becomes a circulant matrix. The other simply pads L trailing zeros to each block at the transmitter, giving rise to a Toeplitz channel matrix.

Since the delayed multipath components of symbols will be separated by the *circular* autocorrelation properties of the spreading sequences in (3), as will be detailed later, the circulant channel matrix is preferable. Therefore, the CP approach is adopted. After the CP insertion and removal on a block-by-block basis, the equivalent input/output (I/O) relationship is given by

$$\mathbf{y}(i) = \tilde{\mathbf{H}}(i)\mathbf{x}(i) + \mathbf{z}(i) \quad (8)$$

where $\tilde{\mathbf{H}}(i)$ is the equivalent circulant channel matrix with the first column $[h(i; 0), \dots, h(i; L), \mathbf{0}_{1 \times (M-L-1)}]^T$. Utilizing the circular shift matrix in (2), it can be reexpressed as

$$\tilde{\mathbf{H}}(i) = \sum_{l=0}^L h(i; l) \mathbf{T}^l. \quad (9)$$

With this block-wise equivalent I/O, the block index i will be dropped in the rest of the paper for notational brevity.

C. Receiver Processing

Substituting (5) and (9) into (8), as the equivalent circulant channel matrix is formulated, the discrete convolution is transformed to matrix multiplication and one can obtain the received signal block as

$$\begin{aligned} \mathbf{y} &= \sum_{l=0}^L h(l) \mathbf{T}^l \sum_{j=0}^{J-1} s(j) \mathbf{T}^{j(L+1)} \mathbf{c} + \mathbf{z} \\ &= \sum_{j=0}^{J-1} s(j) \sum_{l=0}^L h(l) \mathbf{T}^{j(L+1)+l} \mathbf{c} + \mathbf{z}. \end{aligned} \quad (10)$$

In (10), for each symbol $s(j)$, the I/O relationship is transformed from one where a single sequence experiences a circulant multipath channel to one where $(L+1)$ superimposed circularly shifted sequences, each being multiplied by a single channel tap. For all J symbols, $J(L+1)$ different circularly shifted sequences superimposed are obtained, which can be readily separated by the circular autocorrelation properties of the applied multiple shift sequences in (3).

The received signal block in (10) contains independent components, the payload signal $s(j)$, single channel tap $h(l)$ and the shifted versions

of the original spreading sequence $\mathbf{T}^{j(L+1)+l} \mathbf{c}$, by which motivation, a matched filter in the same form of $\mathbf{T}^{j(L+1)+l} \mathbf{c}$ is designed, that is the original spreading sequence $\mathbf{c}$ circularly shifted by $j(L+1)+l$ chips. Multiplying $\mathbf{y}$ by $[\mathbf{T}^{j(L+1)+l} \mathbf{c}]^T$, items $s(j)$ and $h(l)$ can be extracted. The matched-filtering output can be expressed as

$$\begin{aligned} v(j; l) &= [\mathbf{T}^{j(L+1)+l} \mathbf{c}]^T \mathbf{y} \\ &= [\mathbf{T}^{j(L+1)+l} \mathbf{c}]^T \sum_{j'=0}^{J-1} s(j') \sum_{l'=0}^L h(l') \mathbf{T}^{j'(L+1)+l'} \mathbf{c} \\ &\quad + [\mathbf{T}^{j(L+1)+l} \mathbf{c}]^T \mathbf{z}. \end{aligned} \quad (11)$$

Notice the following properties of the circular shift matrix $\mathbf{T}$

$$\mathbf{T}^T = \begin{bmatrix} \mathbf{0}_{(M-1) \times 1} & \mathbf{I}_{M-1} \\ 1 & \mathbf{0}_{1 \times (M-1)} \end{bmatrix} = \mathbf{T}^{M-1} \quad (12)$$

$\mathbf{T}^T \cdot \mathbf{T} = \mathbf{T}^{M-1} \cdot \mathbf{T} = \mathbf{T}^M = \mathbf{I}_M$ and $[\mathbf{T}^i]^T = \mathbf{T}^{M-i}$. Therefore, (11) could be further derived as

$$\begin{aligned} v(j; l) &= [\mathbf{T}^{j(L+1)+l} \mathbf{c}]^T \sum_{j'=0}^{J-1} s(j') \sum_{l'=0}^L h(l') \mathbf{T}^{j'(L+1)+l'} \mathbf{c} \\ &\quad + [\mathbf{T}^{j(L+1)+l} \mathbf{c}]^T \mathbf{z} \\ &= \sum_{j'=0}^{J-1} s(j') \sum_{l'=0}^L h(l') \mathbf{c}^T [\mathbf{T}^{j(L+1)+l}]^T \mathbf{T}^{j'(L+1)+l'} \mathbf{c} \\ &\quad + [\mathbf{T}^{j(L+1)+l} \mathbf{c}]^T \mathbf{z} \\ &= \sum_{j'=0}^{J-1} s(j') \sum_{l'=0}^L h(l') \mathbf{c}^T \mathbf{T}^{[M-j(L+1)-l+j'(L+1)+l']} \mathbf{c} \\ &\quad + [\mathbf{T}^{j(L+1)+l} \mathbf{c}]^T \mathbf{z} \\ &= Mh(l)s(j) - v_I(j; l) + \eta(j; l), \quad l = 0, \dots, L. \end{aligned} \quad (13)$$

The first term in (13) is derived when $l' = l$ and $j' = j$

$$Mh(l)s(j) = \sum_{j'=0}^{J-1} s(j') \sum_{l'=0}^L h(l') \mathbf{c}^T \mathbf{T}^M \mathbf{c} \quad (14)$$

and the second term in (13) is derived when $l' \neq l$ or $j' \neq j$

$$v_I(j; l) = \sum_{j'=0}^{J-1} \sum_{l'=0}^L h(l') s(j'), \quad l' \neq l \text{ or } j' \neq j. \quad (15)$$

Fig. 1 shows the baseband transceiver diagram. The correlating output in (13) contains three components, $Mh(l)s(j)$ is the main component conveying the desired j th signal weighted by appropriate channel taps at the current block, $v_I(j; l)$ is the interference introduced by the intersymbol from multipath delay spread when $j' = j$ and cosymbol from the other $J-1$ symbols superimposed simultaneously when $j' \neq j$, $\eta(j; l)$ follows the the distribution of $\mathcal{CN}(0, M\sigma^2)$. To overcome channel variation, one superimposed symbol [say $s(0)$] is used as the pilot for channel estimation at the current block

$$\hat{h}(l) = \frac{v(0; l)}{Ms(0)}, \quad l = 0, 1, \dots, L. \quad (16)$$

The mechanism of channel estimation is given by Fig. 2. This estimated channel can be then used to demodulate the $(J-1)$ data symbols by collecting the energy of all channel taps as follows:

$$\hat{s}(j) = \frac{\sum_{l=0}^L \hat{h}^*(l) v(j; l)}{M \sum_{l=0}^L |\hat{h}(l)|^2} = \frac{\sum_{l=0}^L \hat{h}^*(l) v(j; l)}{M \|\hat{\mathbf{h}}\|^2}. \quad (17)$$

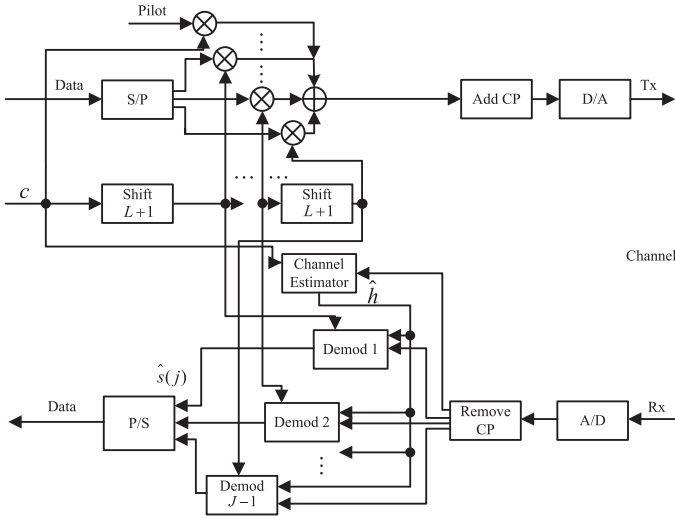


Fig. 1. Baseband transceiver diagram. At the transmitter, multiple symbols are modulated on different circularly shifted versions of multiple sequences. At the receiver, it is an original spreading sequence c shifted by $L + 1$ chips circularly each j time serving as a matched filter.

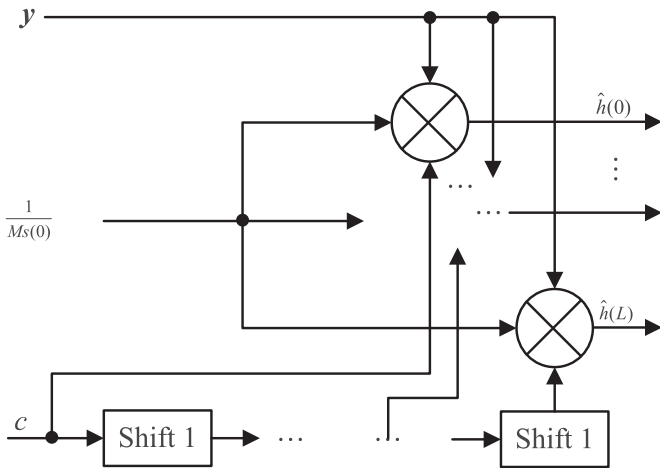


Fig. 2. Channel estimation block in Fig. 1. Shift the first superimposed spread waveform c by 1 bit circularly ($L + 1$ circular times) and project the circularly shifted spread waveforms onto received signal y individually, each output finger is divided by $M s(0)$, channel taps can be estimated.

where $M \sum_{l=0}^L |\hat{h}(l)|^2 = M \|\hat{\mathbf{h}}\|^2$. Fig. 3 shows that the estimated channel taps are projected onto matched-filtering output fingers individually to collect symbol energy.

In the derivations above, the pilot and data symbols ride on the sequences within the same block, i.e., they experience exactly the same channel delay spread during the current block, even in the presence of channel variation. This process ensures the reliability against moderate channel variation, as long as it is not severe enough to ruin the matched-filtering output in (13).

D. Performance Analysis

For a general UWA channel with delay spread $L > 1$, a spreading sequence with good periodic correlation properties is preferable in the absence of channel statistic characteristics at the transmitter end. The periodic autocorrelation function of the spreading sequences with lags

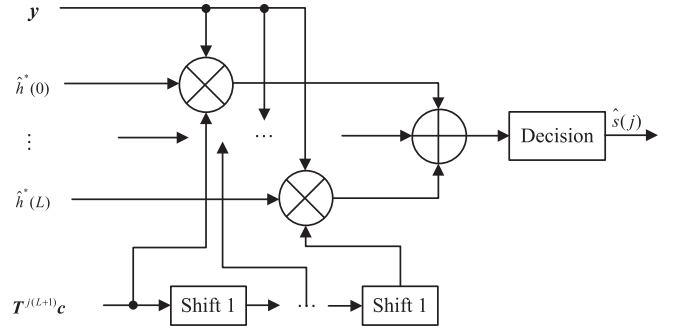


Fig. 3. j th demodulation block in Fig. 1. Shift the j th matched filter $T^{j(L+1)}c$ by 1 bit circularly ($L + 1$ circular times) and project them onto the received signal y , $L + 1$ output fingers can be extracted. Estimated channel taps in Fig. 2 are projected onto output fingers individually, the j th symbol energy is collected.

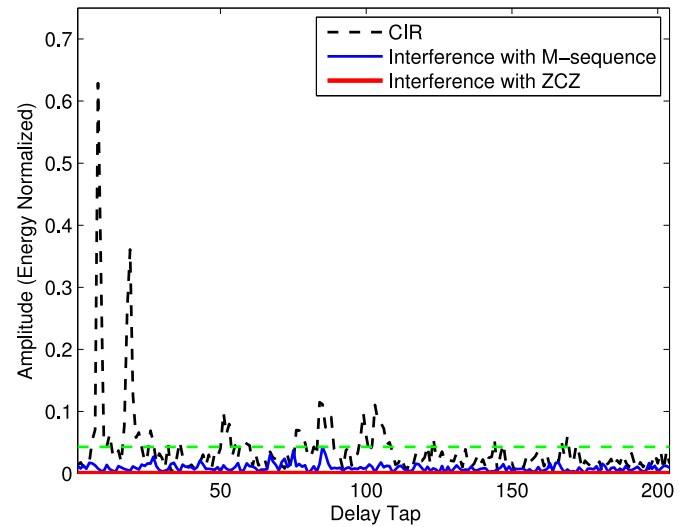


Fig. 4. One snapshot of channel in Zhoushan15 denoted by black dashed curve, delay spread $L = 204$. The sequence length is 1023 for m-sequence and 1024 for ZCZ, two symbols superimposed, respectively. Blue and red solid curves represent interferences by a correlation output for actual data transmission in this snapshot.

of interest would be

$$\begin{aligned} \tilde{\mathbf{r}}_m &= \mathbf{c}^T \mathbf{T}^m \mathbf{c} \\ &= \begin{cases} M, & (m \bmod M) = 0, \\ 0, & (m \bmod M) \in [1, L] \cup [M - L, M - 1] \end{cases} \quad (18) \end{aligned}$$

assuming that $M > 2L$. The periodic autocorrelation function $\tilde{\mathbf{r}}_m$ over $(m \bmod M) \in [L + 1, M - L - 1]$ has no impact on the correlation performance and aims to suppress the correlations over only the lags of interest within $(m \bmod M) \in [1, L] \cup [M - L, M - 1]$.

Such a spreading sequence with periodic correlation region in (18) can eliminate the interferences $v_I(j; l)$, leading to uncorrelated $\eta(j; l)$, and simplifying (13) to

$$v(j; l) = M h(l) s(j) + \eta(j; l), \quad l = 0, \dots, L. \quad (19)$$

Via (19), an L -tap multipath channel decomposes efficiently into L parallel independent channel taps, realizing freedom of interference caused by multipath and multiple symbols superimposed simultaneously across the correlating output fingers; the estimated

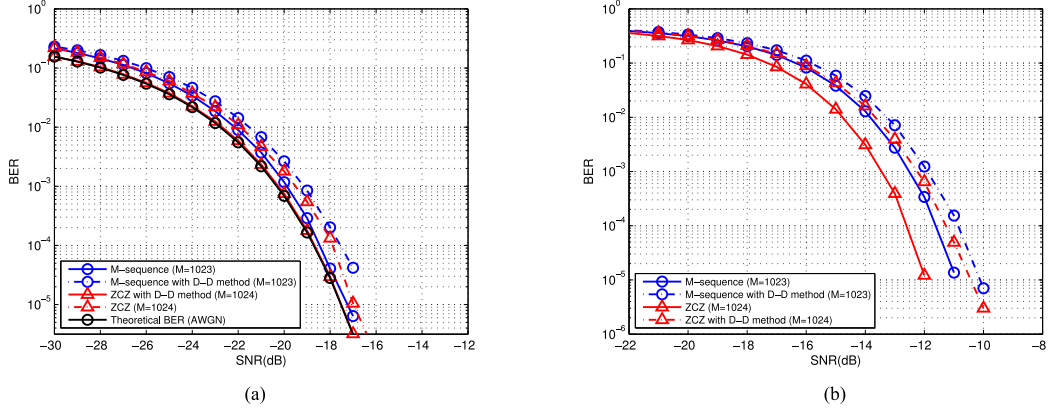


Fig. 5. Simulation results. (a) Performance comparison of the proposed method under static multipath channel with theoretical BER in AWGN channel. (b) Performance comparison of the proposed scheme with D-D DSSS scheme under imperfect CSI.

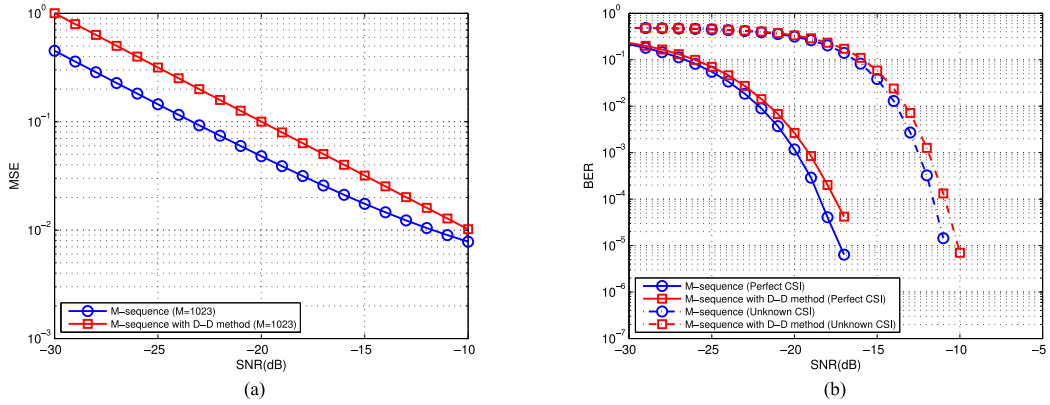


Fig. 6. Performance sensitivity to channel estimation error. (a) Channel estimation MSE versus SNR. (b) BER performance with perfect CSI and unknown CSI.

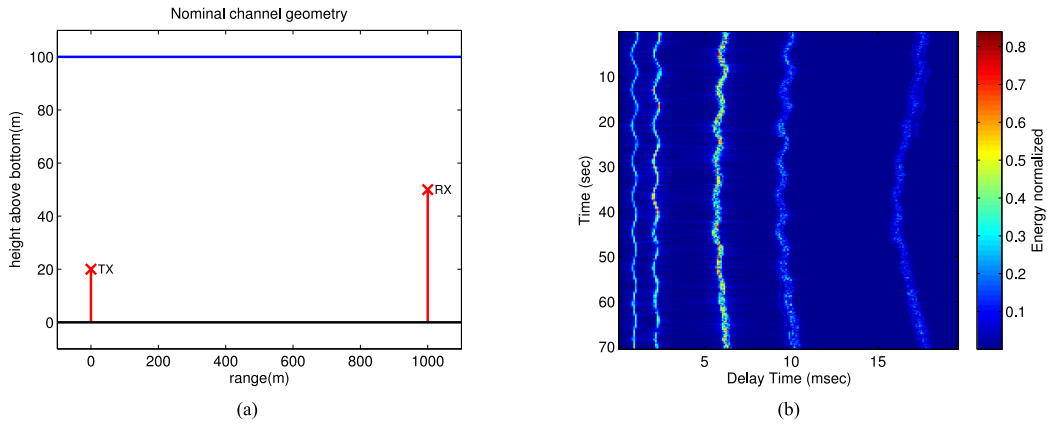


Fig. 7. Time-variant UWA channels. (a) Simulation setup. (b) Normalized CIR evolution over approximately 70-s period.

symbol is given by (assuming a perfect channel estimation, i.e., $\mathbf{h} = \hat{\mathbf{h}}$)

$$\hat{s}(j) = \frac{\sum_{l=0}^L \hat{h}^*(l)v(j;l)}{M\|\hat{\mathbf{h}}\|^2} = s(j) + \frac{\sum_{l=0}^L \hat{h}^*(l)\eta(j;l)}{M\|\hat{\mathbf{h}}\|^2}. \quad (20)$$

Of the two algorithms considered, ZCZ aims to suppress the correlations totally over only the lags of interest, leading to interference-free transmission, i.e., $\hat{r}_m = 0$, $(m \bmod M) \in [-M/4, -1] \cup [1, M/4]$. However, the shift region is reduced from M to $\lfloor M/2 \rfloor$ and accord-

ingly $J_{\max}^{ZCZ} = \lfloor M/2(L+1) \rfloor$, the data rate is reduced by at least 50%. As for m-sequences, the interference is controllable reaching a lower bound. The appendix derives a closed form of SIR. Given the received signal after a simple matched filter, SIR for any symbol is lower bounded by the spreading sequence length M when J equals $J_{\max}$ in (6); this lower bound is very tight. To illustrate the level of interference suppression over extended multipath arrivals Fig. 4 shows the interferences from the periodic correlating output compared with true channel snapshot. It also shows that the interference is

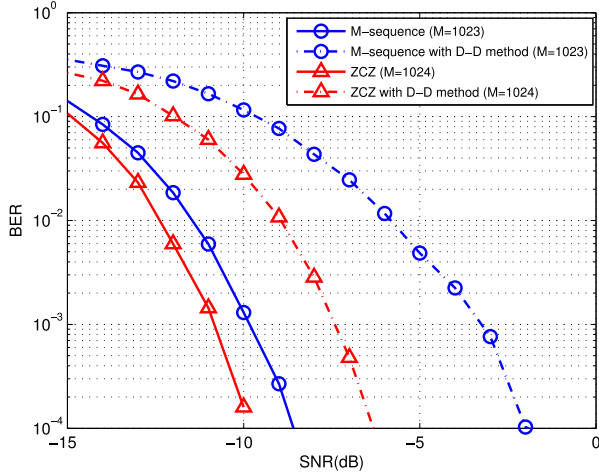


Fig. 8. Simulation results. Performance comparison of the proposed method with D-D method in time-variant channel.

controllable with m-sequence while free with ZCZ over lags of interest. Since symbol acquisition requires integrating the CIR over the multipaths, the amount of interference increases with the length of the multipath spread.

For practical UWA communications, either channel taps $L + 1$ cannot be acquired beforehand or the relationship $M > 2L$ does not always hold; to combat the obstacle partially under the situation of no orthogonality destruction in spreading waveforms, M can be increased. For a typical UWA DSSS system with 10 kc/s that uses a spreading sequence consisting of 1000 chips approximately, the sequence block duration is about 100 ms; if the multipath delay spread of a shallow water channel is on the order of 10 ms [10], $L = 100$, the block fading assumption still makes sense. Admittedly, if the multipath delay spread spans over half of the block length, i.e., $L > M/2$, there is degradation in performance. Thereby, $L < M/2$ is still assumed, and the superimposed operation can be implemented.

III. SIMULATIONS

Considering a DSSS system with signal bandwidth $B = 10$ kHz, the spreading sequence length $M = 1023$ for m-sequence and $M = 1024$ for ZCZ, the chip interval $T_c = 1/B = 0.1$ ms, and the symbol interval is $T_{sym} = MT_c \approx 100$ ms. For a channel with delay spread $\tau_{max} = 20.4$ ms, the channel taps are $L = 204$, one modulates $J_{max} = 5$ for m-sequence (five symbols/sequence duration) and $J_{max} = 2$ for ZCZ as shown in (6). The channel estimation is on chip level and the channel gain is normalized as $\mathbb{E} \left\{ \sum_{l=0}^L |h(i; l)|^2 \right\} = 1$. The motivation for utilizing DSSS is the high reliability under low input-SNR. Hence, BER performance is verified at low SNR, which is defined as E_{chip}/E_0 accordingly and its relationship to E_b/E_0 (per information bit basis) is

$$(E_{chip}/E_0)_{dB} = (E_b/E_0)_{dB} - 10\log_{10}(M). \quad (21)$$

A. Case I: Static Multipath Channel

Assuming a static multipath channel

$$h(t, \tau) = \sum_{l=0}^L \mathcal{A}_l \delta(\tau - \tau_l) \quad (22)$$

where $\mathcal{A}_l$ and τ_l are the amplitude and delay of the l th tap, respectively. The amplitude-delay profile is obtained from one channel snapshot of Zhoushan15 sea experiment, as given in Fig. 4. The channel intensity profile is selected as $\phi(l) = \exp(-0.1l)$ and the total energy is nor-

malized to 1; all the channel phases are i.i.d. complex-valued Gaussian process.

For QPSK modulation, the BER associated with the optimum receiver structure in AWGN channel is given by [25]

$$P_b = \frac{1}{2} \text{erfc} \left(\sqrt{\frac{M \cdot \text{SNR}}{2}} \right) \quad (23)$$

where $\text{erfc}(\cdot)$ represents the complementary error function. The proposed method under static multipath channel is compared with optimum receiver under AWGN channel, the results are given in Fig. 5(a). Simulation results show that the proposed method with ZCZ superimposed matches exactly with the theoretical BER in AWGN channel, attributing to the perfect periodic correlation characteristics. Also the gap between ZCZ and m-sequence is approximately 0.5 dB; performance degradation for m-sequence is owing to the residual interferences induced by intersymbol and cosymbol.

In most situations, channel state information (CSI) is imperfect, to provide convincing results, BER of the proposed scheme is compared with D-D DSSS scheme, as shown in Fig. 5(b). The performance degrades when UWA channels need to be estimated. Meanwhile, the proposed method still works well at low SNRs; BER reaches 10^{-4} at -12.5 dB with ZCZ superimposed, and the gap between the proposed method and D-D DSSS scheme is approximately 1 dB at BER of 10^{-4} .

In practical scenarios, the performance of such systems will be saturated due to the nonorthogonality induced by the different multipath delays. SIR for any symbol is lower bounded by M , as given in the appendix, i.e., $10\log_{10}(M)$ in decibel. For example, $M = 1024$, $\text{SIR} \approx 30$ dB, in which case, error floor may show when E_b/E_0 closes to 30 dB. BER in Fig. 5(b) reaches 10^{-5} when SNR in E_{chip}/E_0 is -12 dB for ZCZ, i.e., E_b/E_0 is 18 dB, which is far from 30 dB. Therefore, error floor does not show.

Imperfect CSI degrades the receiver performance. Recalling formula (13), signal recovery is affected by the precision of channel estimation and noise level. The proposed method utilizes the first superimposed signal $s(0)$ as the pilot to estimate the channel but random errors remain. The mean square error (MSE) of channel estimation is defined as

$$\text{MSE} = \frac{\mathbb{E} \left[\sum_{l=0}^L |\hat{h}(l) - h(l)|^2 \right]}{L + 1}. \quad (24)$$

Fig. 6(a) demonstrates MSE versus SNR between the proposed method and the D-D DSSS scheme. Fig. 6(b) reveals performance sensitivity to imperfect CSI, when SNR varies from -30 to -22 dB; descending tendency of BER performance with our proposed scheme remains slow due to the obstacle of channel estimation and symbol synchronization at low input-SNR.

B. Case II: Time-Variant Channel With Doppler Scales

Consider a time-varying channel where all the paths have different delays and doppler scales, derived as

$$h(t, \tau) = \sum_{l=0}^L \mathcal{A}_l(t) \delta(\tau - \tau_l(t)) \quad (25)$$

where $\mathcal{A}_l(t)$ and $\tau_l(t)$ are the amplitude and delay, respectively, of the l th tap, continuously time-varying due to motion of the transceiver platforms, reflection off the moving surface, refraction in the fluctuating propagation medium, etc. Assuming the delay variation is approximately linear with time during each block duration, a first-order polynomial fitting is

$$\tau_l(t) \approx \tau_l - \alpha_l t \quad (26)$$

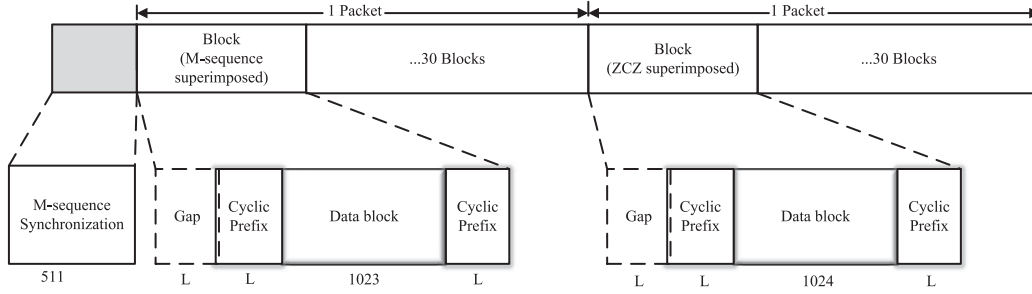


Fig. 9. Transmitted packet structure.

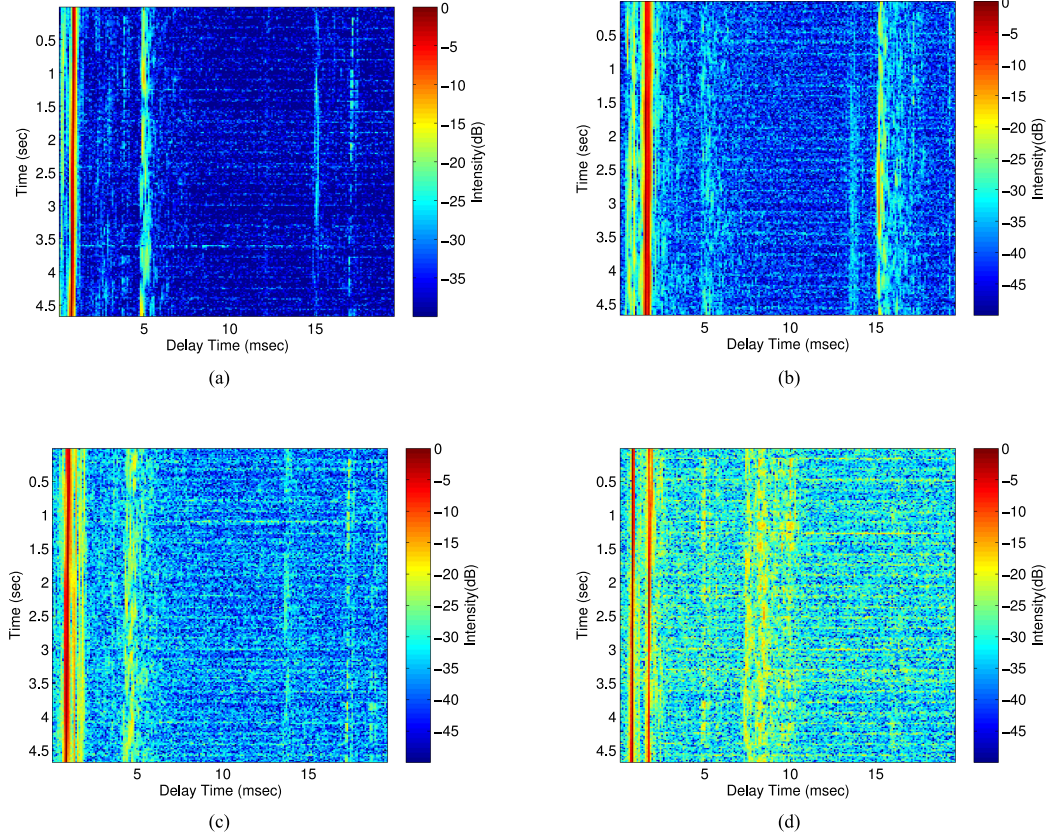


Fig. 10. Measured CIRs normalized within one packet length 4.5 s approximately using calm sea data. The modulus of the channel tap is shown in decibel. (a) and (b) Benign channel conditions at the range of 500 m. (c) Benign channel conditions at the range of 1000 m. (d) Benign channel conditions at the range of 1500 m.

where τ_l is the delay at the start of the spreading block, i.e., the initial delay at $t = 0$, and α_l is often termed as Doppler scaling factor

$$\alpha_l = v_l / c \quad (27)$$

where v_l is the relative platform motion at l th tap and c is the sound speed set as 1500 m/s. With such approximations of delay variation, the time-variant CIR in (25) can be simplified as

$$h(t, \tau) = \sum_{l=0}^L \mathcal{A}_l(t) \delta(\tau - [\tau_l - (v_l/c)t]). \quad (28)$$

To make the simulation more convincing, the simulation environment is introduced by [26]; the ocean depth is 100 m as shown in Fig. 7(a).

The acoustic source is at the depth of 20 m above the bottom and the receiver is 50 m above the bottom, and the range is 1000 m. Due to the motion of the platforms as well as the medium variability and fluctuating reflection boundaries, the numerical model simulates a doubly dispersive channel with large delay spread and severe Doppler spread, as shown in Fig. 7(b).

Assuming the transmitted signals experience time-variant channels possessing imperfect CSI, the superimposed spreading sequence conveying the intended symbols convolves channel taps which update on chip level each operation, named as time-variant convolution. The BER curves within the proposed method and D-D DSSS scheme are depicted in Fig. 8. The proposed method with ZCZ superimposed outperforms the D-D method by 4 dB at BER of 10^{-3} , while 7 dB at BER of 10^{-3} with m-sequence superimposed. In time-variant UWA channels, m-sequence cannot accommodate the rapid parameter variation; nonorthogonality is induced by the doubly dispersive phenomenon. As for ZCZ sequence, although the coherent matched-filtering based receiver cannot combat severe ISI and Doppler shift absolutely, the interferences can be alleviated by the perfect periodic correlations of spreading waveforms.

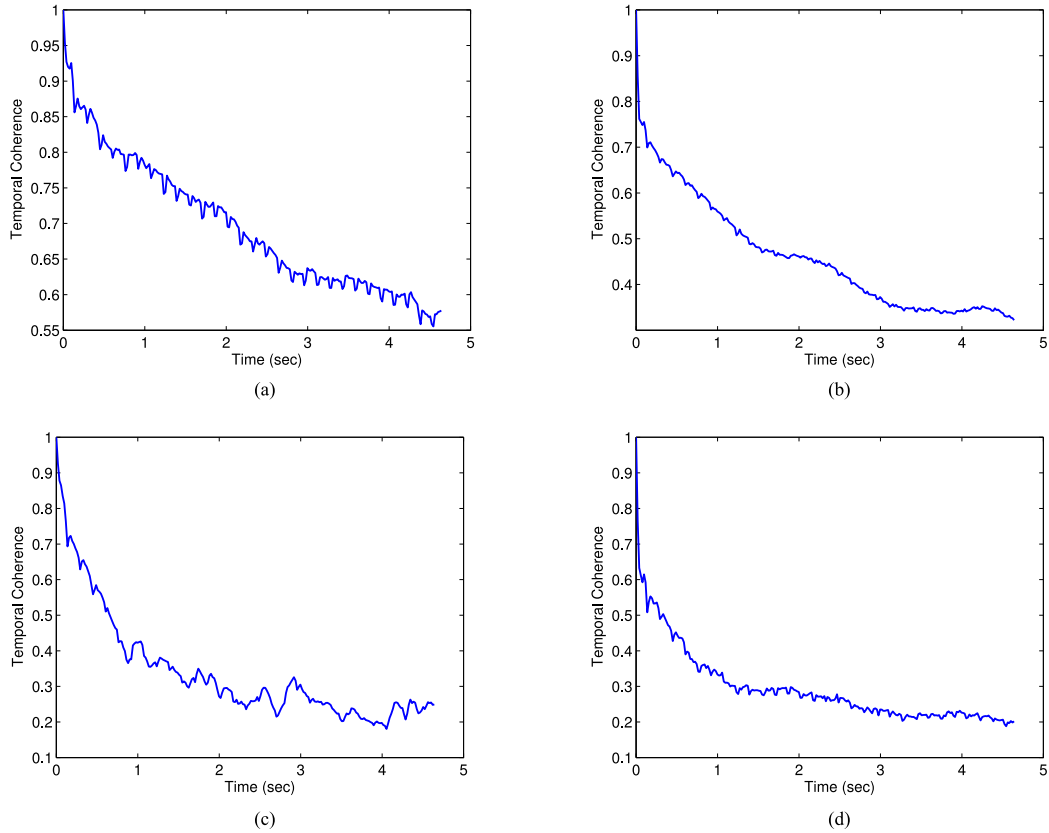


Fig. 11. Intrapacket temporal coherence as a function of lag time under calm sea, corresponding to Fig. 10 individually. (a) and (b) Benign channel conditions at the range of 500 m. (c) Benign channel conditions at the range of 1000 m. (d) Benign channel conditions at the range of 1500 m.

IV. PERFORMANCE RESULTS WITH EXPERIMENTAL DATA

A. Signal Configuration and Experiment

To verify the feasibility of the proposed scheme, a sea experiment was held in the area of Zhairuoshan Island, Zhoushan, Zhejiang, China, on July 20, 2015, at a water depth of 40–60 m. One transducer was deployed at a depth of 20 m submerged from the transmitting ship for transmission, the receiver consisted of four-element vertical hydrophone array, the first one was submerged from the receiving ship to a depth of 20 m, and the array elements were equally spaced at $d = 0.2$ m.

Passband signals were generated at a sampling rate of 250 kHz, the carrier frequency and bandwidth were $f_c = 25$ kHz and $B = 10.4$ kHz, respectively. Pulse shaping was performed by coupling a square-root raised-cosine pulse with a roll-off factor of 0.22. Transmission was organized sequentially, with 30 blocks in each packet, separated by 4.5 s approximately. The DSSS communication system employs the signaling packet illustrated in Fig. 9. Spreading sequence length $M = 1023$ was used for the m-sequence and $M = 1024$ for ZCZ, with chip rate $R_c = 10.4$ kc/s, and symbol duration was $T_{\text{sym}} = MT_c = 98.5$ ms. We choose the CP length and the phase shift step of the spreading sequence as $L = 204$, i.e., $T_{\text{cp}} = T_{\text{gap}} = 19.6$ ms, so that each sequence duration contains $J = 5$ symbols superimposed for m-sequences. One symbol is used as pilot for channel estimation and the other four carry data. With QPSK modulation, $Q = 4$, the data rate is

$$R_b = \frac{J \log_2 Q}{2T_{\text{cp}} + T_{\text{sym}} + T_{\text{gap}}} \approx 63.6 \text{ b/s.} \quad (29)$$

As for ZCZ sequence, each sequence duration contains $J = 2$ symbols superimposed, one symbol is used as pilot for channel estimation and the other one carries data. With QPSK modulation, the data rate is 25.4 b/s.

The SNR of the original received data is around 20 to 30 dB. To generalize the proposed method at low SNR, noise truncated from the oceanic data file is added artificially to the received data for a variable SNR. With different channel conditions, BER performance differs.

B. Channel Characteristics

Considering separated receiver configurations deployed from 500 to 1500 m, the experimental measurements analyzed in this section were recorded at different times and different sea areas on July 20, 2015. Different types of sampled data are processed to assess the impact of channel conditions on the performance of UWA DSSS communications.

1) *Multipath Channel Nearly Constant*: When channel condition is benign, the multipath channel in (22) characterizes the delay variation through a fixed order. These types of channels are plotted in Fig. 10.

To characterize the channel variation over time, one calculates the channel temporal coherence function defined by [13]

$$\rho(\tau) = \frac{[p^*(t)p(t+\tau)]}{\sqrt{[p^*(t)p(t)][p^*(t+\tau)p(t+\tau)]}} \quad (30)$$

where $p(t)$ and $p(t+\tau)$ are the impulse responses of the two neighboring consecutive spreading sequences separated by a lag time τ , t is the sample time of the impulse response. The square bracket $[ab]$

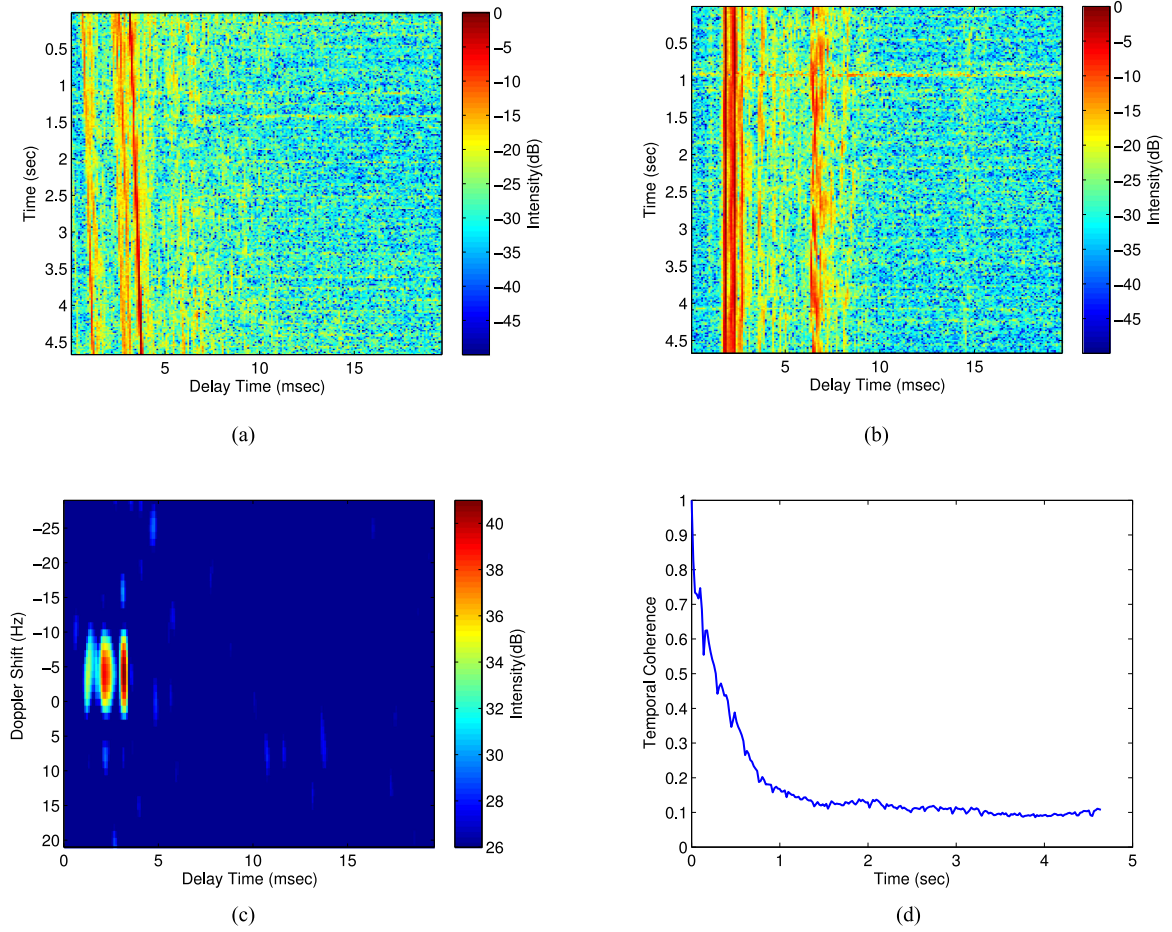


Fig. 12. Measured CIRs normalized within one packet length approximately 4.5 s using rough sea data at the range of 1000 m, the modulus of the channel tap is shown in decibel. (a) and (b) Rapidly time-variant channels. (c) Time-variant scattering function of (a), Doppler scales contain 5-Hz shift and 10-Hz spread. (d) Intrapacket temporal coherence as a function of lag time under rough sea corresponding to (b).

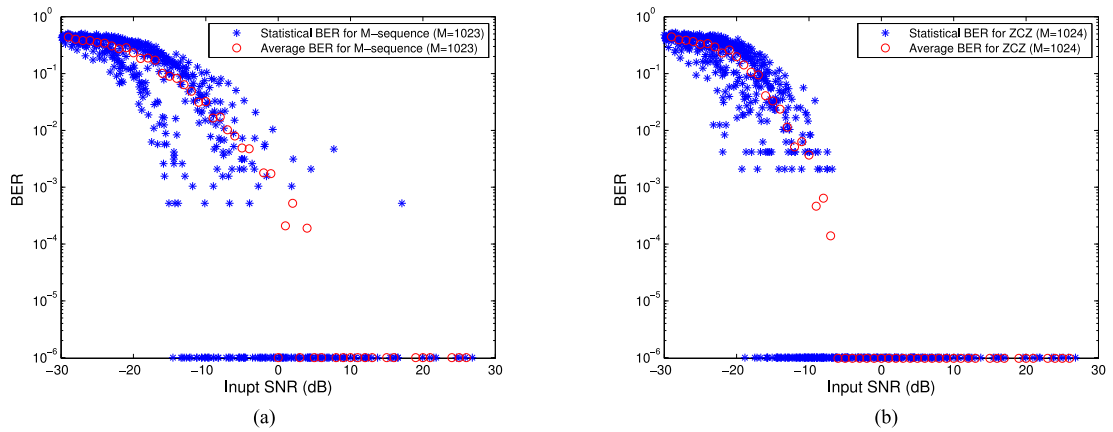


Fig. 13. BER of 304 packets as a function of input-SNR with the proposed method. 10^{-6} is approximately equal to 0 BER on the logarithmic scale, the blue asterisk represents statistical BER and red circle for average BER. (a) M-sequence superimposed. (b) ZCZ superimposed.

denotes the maximum absolute value of the correlation between $a(t)$ and $b(t)$.

The signal coherence time $\tau_{0.8}$ is defined as the lag time when the coherence value drops from 1 to 0.8. One finds in Fig. 11(a) that the temporal coherence time $\tau_{0.8} = 0.6$ s. The data block length is approx-

imately 100 ms. Hence, the channel remains time-invariant within a sequence block.

2) *Doubly Dispersive Channels*: When the channel conditions are severe, all the paths have different delays and Doppler scales, as shown in Fig. 12.

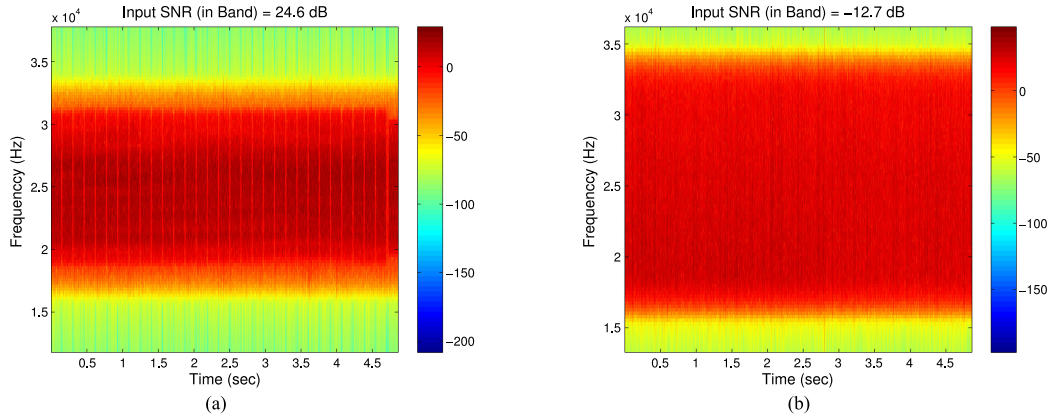


Fig. 14. (a) Spectrum of the received signal with an in-band input-SNR of 24.6 dB occupying a band from 20 to 30 kHz. (b) Spectrum of the received signal with an in-band input-SNR of -12.7 dB and the bandwidth is not visually observable.

At this situation, the receiver performs the following two steps to mitigate the channel Doppler effect.

- 1) The receiver resamples $r(t)$ in the passband to remove the dominant Doppler effect, leading to $y(t) = r(t/(1 + \hat{\alpha}))$, where $\hat{\alpha} = f_d/f_c$ is the resampling factor that corresponds to a rough Doppler estimate.
- 2) The passband signal $y(t)$ is shifted to baseband $y_b(t) = \text{LPF}(y(t)e^{-j2\pi f_c t})$, where $\text{LPF}(\cdot)$ stands for lowpass filtering. Then the receiver performs fine Doppler shift compensation on $y_b(t)$, obtaining $z(t) = y_b(t)e^{-j2\pi \hat{\varepsilon} t}$, where $\hat{\varepsilon}$ is the estimated residual mean Doppler shift.

C. Performance Results and Analysis

Assuming the CSI is unknown *a priori*, the time-variant or time-invariant channel impulse function is estimated via the proposed scheme. To evaluate the performance at low input-SNR, ambient noise data collected during the same experiment are added to reintegrate signals with decreasing SNR, i.e., multiple packets of noise data, randomly chosen, are added to the signal data to generate packets of data with different input-SNRs. There are 304 packets of data for each input-SNR ranging from -30 to 30 dB. Symbol synchronization is done at both high and low SNRs. The resulting uncoded BER is plotted in Fig. 13 as a function of the input-SNR at the receiver. The average BER as a function of input-SNR is also shown in Fig. 13. One finds that, the performance of ZCZ superimposed outperforms m-sequence superimposed, and the average BER is less than 1% for input-SNR ≥ -10 dB; all packets are error free when input-SNR ≥ -6 dB.

Fig. 14 shows the frequency spectrum of the input signal as a function of time (spectrogram); the signal covers a bandwidth from 20 to 30 kHz. When the in-band input-SNR is -12.7 dB, no trace of the received signal in the spectrogram can be detected. Recall the BER performance in Fig. 13; via data equalization, symbols can be decoded accurately.

At low SNR, an unalerted party, not knowing the arrival time of a packet, will be unaware of the presence of the intended signal. It will be difficult for a spectrum-based energy detector (energy integration in the spectrum) to search and detect the signal in blind (LPD). Moreover, it is even unrealistic for a temporal domain energy detector without knowing the signal bandwidth, as the SNR in the time series before passband filter is less than -30 dB (processing gain in this paper is 30 dB). Therefore, the signal will be hard to intercept (LPI) without a prior knowledge of the superimposed spreading sequence information such as sequence length and code types.

V. CONCLUSION

This approach has improved a DSSS modulation based technique, transmitting multiple symbols superimposed simultaneously via the periodic correlation characteristics, with one symbol superimposed as the pilot for channel estimation on chip level. Compared with D-D based DSSS scheme, the proposed approach provides robustness against rapid time-variant UWA channels. By synthesizing spreading sequences with m-sequence and ZCZ, one has shown that the proposed improvement is a generalized structure and possesses good correlation levels. To evaluate the level of interference suppression, a closed form of SIR is given. LPD and LPI properties of the proposed method are discussed. Simulations and experimental data processing demonstrate the validity of the proposed method.

APPENDIX

A quantitative analysis of the interference term in (13) is presented here. Taking m-sequence for example, considering QPSK modulation with $|s(j)| = 1$ and independent taps of the channel with zero mean, given the received signal after a simple matched filter, the SIR in (13) for any symbol is lower bounded by the m-sequence length M . When J equals $J_{\max}$ in (6), this lower bound is very tight. In (13), the first term, the l th delayed multipath component of the j th symbol, is the major component and the second term $v_I(j; l)$ is the interference. Accumulating the signal and the interference energy from all $(L + 1)$ taps in the numerator and the denominator as the definition of SIR in [27], we obtain the SIR for the j th symbol as indicated in (31) shown at the top of the next page.

From (6), we know $J_{\max}(L + 1) \approx M$ and thereby obtain the tight lower bound of SIR as M . In another case, when phases for different rays happen to be the same, namely all the interferences add incoherently, the SIR for the j th symbol is derived as

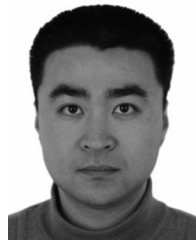
$$\begin{aligned}
 \text{SIR}(j) &= \frac{\mathbb{E}\left\{\sum_{l=0}^L M^2 |h(l)s(j)|^2\right\}}{\mathbb{E}\left\{\sum_{l=0}^L |v_I(j; l)|^2\right\}} \\
 &= \frac{M^2 \sum_{l=0}^L \mathbb{E}\{|h(l)|^2\} \mathbb{E}\{|s(j)|^2\}}{\mathbb{E}\left\{\sum_{l=0}^L \sum_{j=0}^{J-1} \sum_{l'=0}^L h(j; l')^2\right\} \mathbb{E}\{|s(j)|^2\}} \\
 &= \frac{M^2 \sum_{l=0}^L \mathbb{E}\{|h(l)|^2\}}{J^2(L + 1) \sum_{l=0}^L \mathbb{E}\{|h(l)|^2\}} \geq \frac{M}{J_{\max}} \quad (32)
 \end{aligned}$$

$$\begin{aligned}
\text{SIR}(j) &= \frac{\mathbb{E}\left\{\sum_{l=0}^L M^2 |h(l)s(j)|^2\right\}}{\mathbb{E}\left\{\sum_{l=0}^L |v_I(j;l)|^2\right\}} \\
&= \frac{M^2 \sum_{l=0}^L \mathbb{E}\{|h(l)|^2\} \mathbb{E}\{|s(j)|^2\}}{\sum_{l=0}^L \left[\sum_{j'=0}^{J-1} \sum_{l'=0}^L \mathbb{E}\{|h(l')|^2\} \mathbb{E}\{|s(j')|^2\} - \mathbb{E}\{|h(l)|^2\} \mathbb{E}\{|s(j)|^2\}\right]} \\
&= \frac{M^2 \sum_{l=0}^L \mathbb{E}\{|h(l)|^2\}}{[J(L+1)-1] \sum_{l=0}^L \mathbb{E}\{|h(l)|^2\}} = \frac{M^2}{J(L+1)-1} \geq \frac{M^2}{J_{\max}(L+1)-1}.
\end{aligned} \tag{31}$$

where $J_{\max}(L+1) \approx M$ as stated before. In (31) shown at the top of the page, the interference term adds up with different magnitudes and phases, while in (32) all the interferences add up with the same phase. As $h(j;l)$ stands for different propagation path, it is reasonable and practical that every acoustic ray experiences different phase, which is normally uniformly distributed. Therefore, the tight lower bound derived in (31) still makes sense in the scenario in this paper.

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Fengzhong Qu (S'07–M'10–SM'15) received the B.S. and M.S. degrees from Zhejiang University, Hangzhou, China, in 2002 and 2005, respectively, both in electrical engineering. He received the Ph.D. degree in electrical and computer engineering from the Department of Electrical and Computer Engineering, University of Florida, Gainesville, FL, USA, in 2009.

From 2009 to 2010, he was an Adjunct Research Scholar in the Department of Electrical and Computer Engineering, University of Florida. Since 2011, he has been with the Ocean College, Zhejiang University, Zhoushan, China, where he is currently a Professor and an Associate Chair of the Institute of Ocean Sensing and Networking. His current research interests include underwater acoustic communications and networking, wireless communications, signal processing, and intelligent transportation systems.

Dr. Qu is an Associate Editor of the *IEEE TRANSACTIONS ON INTELLIGENT TRANSPORTATION SYSTEMS* and the *IET Communications and China Communications*.



Xiangzhao Qin received the B.S. degree in underwater acoustic engineering from Harbin Engineering University, Harbin, China, in 2015. He is currently working toward the Ph.D. degree in the Ocean College, Zhejiang University, Zhoushan, China.

His research interests include underwater acoustic communications and signal processing.



Liuqing Yang (F'15) received the Ph.D. degree in electrical and computer engineering from the University of Minnesota, MN, USA, in 2004.

She is currently a Professor at Colorado State University, Fort Collins, CO, USA. Her general research interests include signal processing with applications to communications, networking and power systems, subjects on which she has published more than 210 journal and conference papers, three book chapters, and one book.

Dr. Yang received the Best Dissertation Award in the physical sciences and engineering from the University of Minnesota in 2004, the Best Paper Award at the IEEE ICUWB06, ICC13, ITSC14, GLOBECOM14, ICC16 and WCSP16, the ONR YIP Award in 2007, and the NSF CAREER Award in 2009. She was an Associate Editor of several IEEE Transactions, and a TPC Chair/Member for a number of conferences.



T. C. Yang received the Ph.D. degree in high energy physics from the University of Rochester, Rochester, NY, USA, in 1971.

He is currently a Professor and previously a Pao Yu-Kong Chair Professor at Zhejiang University, Hangzhou, China. From 2012 to 2014, he was a National Science Counsel Chair Professor at the National Sun Yat-Sen University, Kaohsiung, Taiwan. Before that, he spent 32 years working at the Naval Research Laboratory, Washington, DC, USA, where he was the Head of the Arctic Section, Dispersive

Wave Guide Effects Group, and the Head of the Acoustic Signal Processing Branch, and consultant to the division on research proposals. His current research interests include environmental impacts on underwater acoustic communications and networking, exploiting the channel physics to characterize and improve performance, environmental acoustic sensing and signal processing using distributed networked sensors, and methods for improved channel tracking and data-based source localization. In earlier years, he pioneered matched mode processing for a vertical line array, and matched-beam processing for a horizontal line array. His other areas of research included geoacoustic inversions, waveguide invariants, effects of internal waves on sound propagation in shallow water, Arctic acoustics, etc.

Prof. Yang is a Fellow of the Acoustical Society of America.